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An integrated TOPSIS and ARAS method multi-criteria decision-making approach for optimizing investment portfolios using goal programming and genetic algorithm model

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As the portfolio optimization field grows, classical techniques often notoriously find it difficult to efficiently model how investors decisions, risk tolerances, and asset attributes intertwine. This paper presents an innovation-based hybrid method, where Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) combined with Additive Ratio Assessment (ARAS) for multi-criteria decision making, Goal Programming (GP) and a Genetic Algorithm (GA) for finding constraints are united. The proposed approach enhances the accuracy of ranking and effectiveness of allocation by incorporating asset evaluation, characterization of investors and probabilistic construction of portfolios. The system is tested in view of various performance implications, using the FAR-Trans dataset, a collection of genuine transaction statistics and asset pricing, as well as investor data. The first step involves project transaction capacities partitioning and risk categorization to create a bipartite TOPSIS–ARAS scoring mechanism. The GP part of the model matches investment decisions to the individual return and risk expectations of each investor, and the GA promotes the use of entropy-aware strategies. Important performance metrics are a Sharpe Ratio of 2.241, the annualized return of 4.6% and diversification score of 0.845. The study also reflects a 0.729 correlation between TOPSIS–ARAS rankings, and GP configurations leading to portfolio returns of over 30.0%. The system offers a realistic depiction of the behavior of investors, considering several transaction channels and different risk factors as well as geographies. The comprehensive integration is very flexible, computationally effective and based on realistic investment models while minimizing constraint deviation.

Keywords Portfolio optimization, Multi-criteria decision-making (MCDM), TOPSIS, ARAS, Goal programming (GP), Genetic algorithm (GA), Investment strategies, Risk management

Background and motivation

The current financial environment calls on investors to operate in volatile markets, deal with complex risks and align their financial objectives with market constraints. Investors today do not only want the highest possible returns. Models that are based on one objective frequently fail to respond to the complexity of the modern investment dilemma, which is why MCDM frameworks are required to optimize portfolios with informed

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choices[1,2]. The TOPSIS and ARAS methods, of all available MCDM frameworks[3,4], have recently gained popularity as options for decision makers. TOPSIS assesses alternatives in relation to how closely they match the ideal and anti-ideal situations and has a powerful discrimination ability, even between assets that are similar in terms of performance[5]. Alternatively, ARAS determines rankings by summing normalized performance ratios, resulting in a strong algorithm for controlling scaling and zero values[6–8].

Conventional portfolio optimization approaches

The integration of TOPSIS and ARAS generates a complete model that drastically enhances the robustness of decision-making processes. TOPSIS is good at understanding faint differences among highly comparable alternatives, whereas ARAS brings stability through additive aggregation. The combination of both techniques results in the reduced sensitivity to normalization factors, reduces the number of rank reversals, and produces a more reliable initial screening of investment portfolios[9–11].

While such MCDM models offer certain advantages, they are primarily confined to the ranking phase and do not directly contribute to asset allocation decisions. In the realm of high-stakes portfolio construction, merely identifying the top-performing alternatives is inadequate; it is essential to determine the optimal allocation levels that satisfy a range of investor-specific constraints. This necessity calls for the application of Goal Programming (GP), a mathematical method that models trade-offs among multiple financial objectives, in conjunction with Genetic Algorithms (GA), a bio-inspired evolutionary optimization technique proficient in navigating extensive, nonlinear, and high-dimensional investment landscapes[12–14].

In the dynamic field of financial investment, portfolio optimization involves not only maximizing returns but also meeting multiple, often conflicting, investor objectives[15]. These objectives encompass minimizing risk, maintaining liquidity, adhering to regulatory constraints, and aligning with investor capacity and sectoral preferences. As financial markets continue to evolve, so too must the models that guide capital allocation decisions. While traditional portfolio optimization techniques provide a foundational basis, they increasingly fall short in addressing the multidimensional nature of real-world investor behavior[16].

Conventionally, two major approaches have significantly contributed to the discipline. Multi-Criteria Decision-Making (MCDM) structures and solutions from quantitative optimization processes. MCDM methods—such as AHP, TOPSIS, VIKOR, ELECTRE, and ARAS—are widely utilized to rank investment options analyzed on the basis of financial indicators such as return, risk, liquidity, and stability[17–20].

These approaches[21,22] create a clear framework of value judging, including both factual information and the investors' personal views. They continue to be limited in terms of using these frameworks to control real assets primarily because there are no mechanisms for determining how much to invest in each asset or imposing limitations like budgetary constraints or return goals. In contrast, models based purely on optimization, for example, mean–Variance Optimization (Markowitz), Goal Programming (GP), and methods such as Genetic Algorithms (GA) are created to numerically optimize returns in terms of risk[23–25].

Although they can deliver practical results within certain limitations, these models tend to work independently of the investor's particular goals and behavioral inclinations. In the absence of a formalized system of ranking preferences, these models may allocate assets that, while theoretically optimal, do not correspond to what investors actually want or are willing to tolerate in the form of risk.

The gap between preference modeling and portfolio optimization has created ad hoc solutions that struggle to bring together high-quality decision, efficient updating and algorithmic practicality. The segregation of asset choice and portfolio formation in real-life scenarios is likely to produce less than optimal investment plans, particularly in cases where there are multiple objectives which include return, risk management, and diversification.

Proposed hybrid framework

To overcome these hurdles, an integrated modular framework will be proposed in this paper to combine decision-making systems and optimization subjected to constraints. The essence of this approach is that two-tier MCDM model is used that is based on TOPSIS when the distances are measured and ARAS when additive normalization is considered. This is enhanced by a Goal Programming (GP) formulation with optimization according to the particular constraint parameters of the investors combined with a Genetic Algorithm in order to arrive at maximum allocation within the GP defined feasible domain of the solution. This way individual preferences which are established in investment decisions are customized and allocation of resources is streamlined to an established set of goals. The model is tested on the FAR-Trans dataset that involves combining investor transactions, demographics, pricing, and financial instruments.

It is proposed to make an advanced hybrid method suggesting the integration of TOPSIS, ARAS and goal programming and Genetic Algorithms to perform as an optimization technique. This complex model supports the concept of prioritized decision making and efficient resource allocation within the conditions of the practical circumstances. The framework presents an opportunity of combining the qualitative decision making with quantitative optimization that can transcend various market conditions and able to use large sets of data.

Research contribution and scope

Not only is the proposed system as accurate and adjustable as traditional models, or more so, but it also has an adjustable scale with a more understandable design that aligns well with the current digital investment platforms. This research improves the application of multi-objective portfolio optimization[26,27] through the use of preference modeling, rule-based decision logic, and evolutionary computation.

Although isolated applications of MCDM techniques and optimization algorithms show success, the field of investment portfolio optimization lacks a comprehensive methodology that combines subjective preference modeling and computational optimization. Techniques such as TOPSIS that identify relative rankings based on the closeness to a best solution, and ARAS that uses additive normalization to create a balanced scoring system, are constrained in modelling capital allocation or practical limitations without integration. For example, sometimes Goal Programming (GP)[28,29] or Genetic Algorithms (GA) are used in the optimization of investments, but they rarely involve prior decision-preference modeling, which makes them ineffective in dealing with the goals of stakeholders[30,31].

Taking this gap into consideration, there is a critical need for a strong methodology that combines the decision-making skills of TOPSIS and ARAS with the optimization methods of GP and GA[32–35]. The absence of integration results in the inefficient selection process of assets, poorly optimized allocation plans and inadequate adaptability to different multi-objective investment landscapes.

Conventionally, the traditional portfolio optimization models make use of Multi-Criteria Decision-Making (MCDM) tools such as TOPSIS or AHP to assess and rank possible investment options. Conversely, they also use optimisation techniques like Goal Programming or Genetic Algorithms, and in many cases they use only quantitative measures of return and risk in their analysis. Such methods lack the capacity to incorporate opinion of investors that is subjective, consider variable restrictions, and reflect dynamic features of feedback.

Its suggested technique integrates the qualitative and the quantitative analysis into a single decision-support framework. It is possible to assume that one of the starting points is investor segmentation, which leads to the adoption of a two-level TOPSISARAS approach to scoring financial instruments in accordance with the return, risk, and liquidity. This is followed with the imposition of constraints on investor specific investments such as wanted returns and budget limits via a flexible Goal Programming technique coupled with a Genetic Algorithm balancing the entropy and diversification with the optimal allocation. The prevalence of these elements enables the given approach to maximize the accuracy, reactiveness, and employability by the different sort of investors.

On the basis of the findings in the earlier limits of the system shown in Fig. 1, the goal of this research is the development of a modular framework of portfolio optimisation combining the stages harmoniously. This paper proposes the usage of a two-layer MCDM framework that combines ARAS and TOPSIS and thereby introduces enhanced performance of this framework. By combining TOPSIS and ARAS, we obtain a multidimensional and reliable analysis of possible investments based on such factors as return, risk, and liquidity.

A Goal Programming (GP) model is incorporated into the framework, designed to accurately define investor goals and restrictions, so that it would be possible to effectively address multiple objectives at the same time. In order to increase the accuracy of allocation decisions, the GP model is supplemented by a Genetic Algorithm, which explores the whole feasible space in search of optimal or suboptimal asset weight assignments.

Experimental results validate the superiority of the integrated model: decision accuracy, risk-return ratios, and the efficiency of processing are improved as compared to traditional models. In addition, the practical utility of the model is tested against standard financial benchmarks, illustrating the model's ability to adjust to different types of investors and changing market environments.

While related works such as Wu et al[36,37]. and Wang et al[38]. have applied MCDM and optimization models to portfolio selection, this study advances those contributions by integrating complementary MCDM techniques (TOPSIS + ARAS), hybridizing deterministic and evolutionary optimizers (GP + GA), and validating the model using a behavior-driven dataset. The distinction lies not merely in integration but in the sequential structure and investor-specific adaptability of the framework.

Related works

Multi-criteria decision models have become indispensable in portfolio selection because they are needed to address the challenges of pursuing conflicting investment goals simultaneously. The central position of procedures such as the Analytic Hierarchy Process (AHP) in the formation of structured decision-making was then succeeded by the introduction of improved ranking systems like the TOPSIS, in their work[5,39], used an AHP–TOPSIS methodology to rank Colombian stock options according to return, risk, and liquidity.

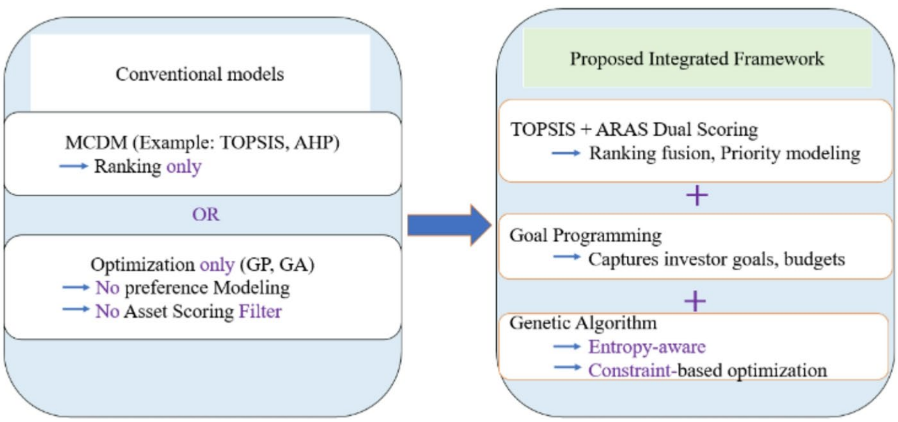

Fig. 1. Overview of the proposed hybrid investment portfolio optimization model which integrates MCDM, Goal programming and genetic algorithm layers.

Moreover, financial scenarios have used different outranking methods, including ELECTRE-TRI and FlowSort, together with AHP and TOPSIS. These models are able to successfully combine investment choices in complex preference frameworks because of the results presented by[40], so offering useful advice in discrete portfolio management.

This topic of research deals with employment and comparison of different Multi-Criteria Decision-Making (MCDM) techniques like TOPSIS, VIKOR and COPRAS in general of Tehran Stock Exchange. The paper shows that MCDM models are valuable flexible appliances in the Finance industry during decision-making. ARAS method has been commended especially because it is an additive assessment method which enables analysts to be able to rate alternatives using standardized scores against best benchmark[8,41,42].

The thing is that the separate use of TOPSIS or ARAS still has limitations. It is notable that TOPSIS is very sensitive to outliers and normalization and ARAS would not be very effective in finding alternatives that are similar to one another. Combining these two approaches will allow the decision-makers to benefit from the complementary properties of these techniques: rank stability can be achieved with ARAS due to their linear utility functions, and TOPSIS is more powerful to differentiate since the distances between points and ideal states are Euclidean. This systematic combination of appraisal each neutralizes the drawbacks of the other two approaches and strengthens the entire assessment process of making critical decisions at the asset-filtering level[43,44].

Resource allocation or optimization can be considered as one of the important phases of investment planning because Multi-Criteria Decision-Making (MCDM) methods[45] may facilitate the selection and rankings however they do not provide the actual stage of resource allocation. Given the modeled multiple soft and hard constraints (return thresholds, risk caps, and so on liquidity quotas), GP provides a powerful framework to model them. Chopra and Chopra (2005) have shown the usefulness of GP in the process of fitting the projects portfolio goals[46], by making minimum deviations towards the set priorities. It has turned out to be a strong metaheuristic that can explore vast and non-linear financial landscapes, the benefits of which have use been applied in asset allocation and risk diversification[12].

To enhance adaptability in uncertain financial environments[47,48], introduced a hybrid decision-making model that integrates fuzzy MCDM[49,50] with multi-objective mathematical optimization. This approach enables decision-makers to account for interval uncertainty in input data, thereby improving the robustness of portfolio selection when precise values are either unavailable or volatile. This contribution is particularly pertinent in capital markets, where ambiguity in forecasting and the evaluation of subjective criteria often impede decision consistency.

While individual methodologies have achieved a certain level of maturity, a significant gap remains in the literature, as only a limited number of studies have proposed a comprehensive framework that integrates Multi-Criteria Decision-Making (MCDM) with both Goal Programming (GP) and Genetic Algorithms (GA). Existing approaches often conclude after the decision-ranking phase or proceed with optimization using predetermined weights, without incorporating preference modeling. Although[51] introduced a two-phase MCDM plus optimization model, it did not integrate complementary MCDM methods (e.g., TOPSIS and ARAS) and failed to incorporate both deterministic (GP) and stochastic (GA) optimization techniques.

In a recent study[51], utilized the Non-Dominated Sorting Genetic Algorithm III (NSGA-III) for multi-objective portfolio optimization, focusing on risk-return trade-offs, as well as kurtosis and skewness metrics. Their methodology effectively generated Pareto-optimal fronts for complex investment strategies, outperforming traditional mean–variance models in addressing conflicting financial objectives. However, the study did not incorporate a structured pre-optimization filtering mechanism using preference-based models, such as MCDM, which led to an indiscriminate search across the entire portfolio space without strategic prioritization. This underscores the need for a hybridized approach, wherein qualitative evaluation methods, such as TOPSIS and ARAS[52,53], can serve as an initial screening layer, followed by NSGA-III or GA-based optimization to refine allocation decisions.

This study addresses a critical gap in the literature by introducing a comprehensive hybrid model. The suggested approach begins by taking an integrated TOPSIS-ARAS MCDM layer to filter the portfolios after which this outcome is used as an input to a Goal Programming module that allows taking into consideration the objectives, which exist and form of the investor. The final optimization is executed through Genetic Algorithms. This three-tiered approach facilitates scalable, intelligent, and preference-aligned portfolio optimization, representing a significant advancement over existing fragmented models.

Methodology and mathematical formulation

Figure 2 presents a modular architecture that employs feedback mechanisms to optimize portfolios with multiple objectives. The process initiates with the acquisition of raw data on investors and assets from the FAR-Trans dataset. Subsequently, the system categorizes investors based on their risk profiles and investment capacities. The pre-processing phase includes Z-score normalization, one-hot encoding of categorical data, and time-based filtering to establish the decision matrix.

Combining the TOPSIS to rank by distance with ARAS to provide ratio score, the MCDM framework acquires the total asset rank within its two-level performance. When assets fail to rank according to the threshold specified, there is a need to alter constraints or reweighting of assets. The Goal Programming (GP) model will be applied to incorporate the goals the investor has specified especially in terms of returns, risk and budget limits on the amount of the assets above the agreed limit. Following this, the Genetic Algorithm (GA) optimization module refines position weights via fitness calculations and introduces operators, which are Simulated Binary Crossover (SBX), mutation, and seek to maximize the return and prioritize the penalties undertaken.

Optimal sector resultant and balanced weight assignment have a big impact on the final assets allocation. These allocations are analyzed with the help of such key performance indicators as Sharpe Ratio, Return on

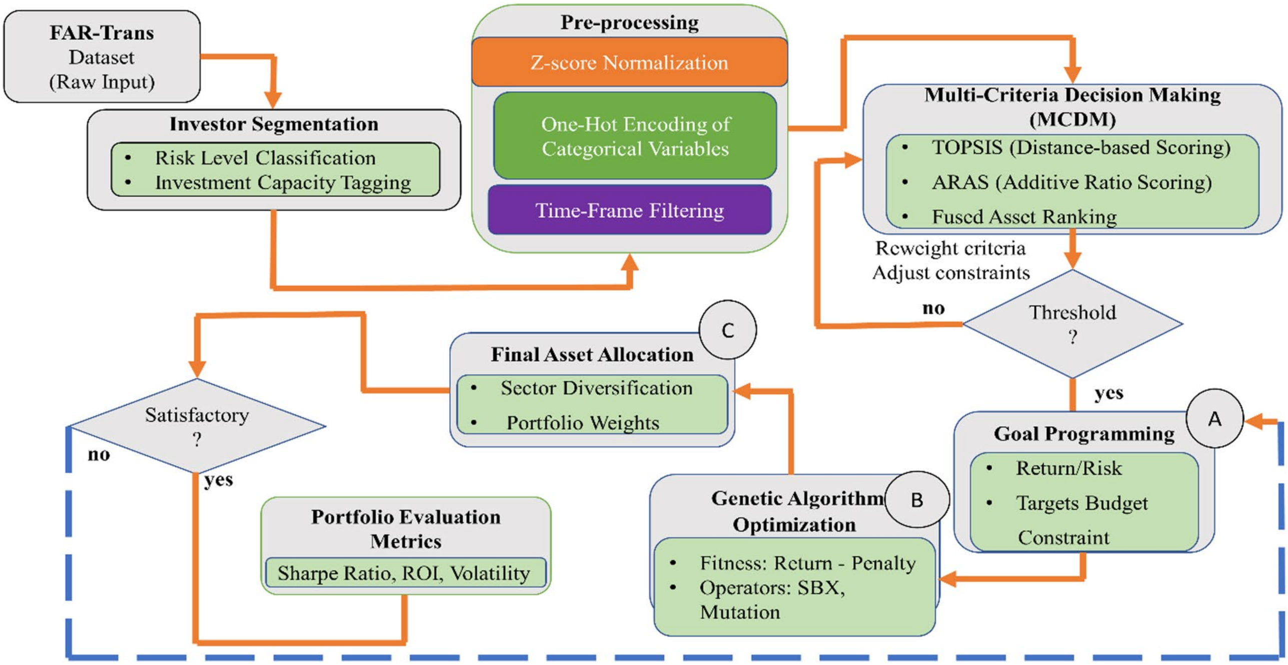

Fig. 2. Workflow of the proposed hybrid MCDM–GP–GA framework for intelligent portfolio Optimization.

Investment (ROI), Volatility. When original results fail to meet standards, the framework offers the modifications via repetition of first steps, i.e. recalculation of MCDM scores or adjustment of constraints, thus, maintaining the closed-loop process. The application of this framework results in agile and investor-oriented portfolios that display versatility and dependability under differing financial circumstances and guidelines.

As shown in Fig. 2, the MCDM scoring layer outputs a ranked list of assets based on investor preferences, which then forms the input set for Goal Programming (GP) and Genetic Algorithm (GA) modules. The flow from behavioral data to optimization is thus preserved across all stages.

Dataset overview and preprocessing

This research presents the FAR-Trans dataset[54], a publicly available dataset that is specifically intended to support research in financial asset recommendation. The Far-Trans dataset includes anonymized retail investor activity, comprehensive tracking of asset prices, and profiles of investors, obtained from a leading European financial institution for the period January 2018 to November 2022. A systematic preprocessing pipeline was designed, which includes elimination of redundancies, standardization of price discrepancies and harmonization of transaction records. Categorical variables were pre-processed with one-hot encoding, and continuous variables were scaled through min–max normalization to be compatible with multiple machine learning techniques. The preference data of investors in the form of risk toleration and investment capacity were obtained to enable specific methods of recommendation. Also, the current research evaluates the effectiveness of eleven algorithms to recommend financial assets on the dataset.

This study is aimed at coming up with a well-matched product suggestion system on the surety of customer finances. The dataset includes customer transaction history and the product details. The categorical columns were converted to one-hot encoded format to allow numeric input to be used in machine learning models. To put continuous features on the same scale, in terms of importance of all the features, minmax scaling was carried out. Moreover, user-item matrix was adapted in the study to explore the preferences and behaviours of users thereby helping in creation of customized recommendation strategies.

This paper is related to the investigation of the accuracy of five machine learning techniques in the matter of financial risk assessment. The data contains different types of financial details, and pertinent customer data. Min–max scaling has been done as an adjuvenation process of the data to normalize the ranges of the independent variables to make the data fit to modeling. The categorical variables were encoded using one-hot encoding to make them friendly to the machine learning. These preprocessing steps ensured that the data could be used in the machine learning models and therefore gave a more equal chance of evaluating the performance of the various approaches in financial risk assessment.

The input data on the MCDM layer was taken out of the FAR-Trans data. These were both categorical and numerical variables, including the rate of returns, the standard deviation, the popularity of assets and the score of investor preferences. The decision matrix $X = [x \ ij \ ]$ with each row corresponding to an alternative i and each column to a criterion j was constructed. Categorical data were treated by one-hot encoding and numerical data by min–max normalization to obtain the range of [0,1]. The TOPSIS and ares based on this process of the standardized matrix in the following section.

Multi-criteria decision-making layer: TOPSIS and ARAS

In recent past, portfolio analysis has diversified in its reach by the adoption of multi-criteria aids that increase the investment selections as well as their orientations favoring the investor interests. In determining stock performance on risk, returns, and liquidity, a meaningful research effort was employed, which involved application of a hierarchical ranking technique in identifying stock performance of the same. The method employed the standardization of financial measures as well as the distance-based assessment to rank investment alternatives, a factor that would assist investors in developing areas to make wiser choices.

Researchers attempted to determine the best strategies to adopt in the selection of stocks using additive scoring and relative closeness as aspects of a comparative analysis. The models were used for the analysis of a matrix of normalized financial indicators and then the stability of rankings and weight correction based on investors' preferences was assessed. It was discovered that there was great match between the ordering done by the models and the historical returns of assets traded in various exchanges.

A later investigation employed a two-staged process that first entailed the evaluation of asset options through a normalized multi-criterion scoring process with even weighting factors. The resulting ranked outputs in turn guided an optimization module, which showed the relevance of a structured pre-selection in terms of reducing computation needs and improving investment performance. Such implementations reflect the increasing adoption of the structured ranking methods in investment decision-making, and, specifically, those involving normalized performance indicators, distance metrics, and additive scoring to match investor-specific priorities.

The model for the integration of TOPSIS and ARAS proposed in this paper brings forward a hybrid scoring system designed to achieve more robust and stable asset rankings. details the methodology of this integration, making the asset ranking process both differentiated and stable, complying with the investors' preferences. To improve methodological transparency and reproducibility, we present the dual-layer MCDM model (TOPSIS and ARAS) with clear step-wise computations and variable definitions. Each equation is aligned with the hybrid scoring mechanism that guides the portfolio filtering stage. The TOPSIS component emphasizes distance from ideal solutions, while ARAS focuses on additive ratios. Their fusion balances differentiation and stability in portfolio ranking.

Step 1 Normalize the Decision Matrix.

For TOPSIS: Use vector normalization

$$r_{ij}^{TOPSIS} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{m} x_{ij}^{2}}} \tag{1}$$

where: x_{ij} is the original score of the i^{th} alternative on the j^{th} criterion, m is the number of alternatives.

Result: Scales each criterion vector-wise for fair comparison.

For ARAS: Normalize using additive method

$$r_{ij}^{ARAS} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}} \tag{2}$$

This method makes each criterion sum to 1, preserving relative proportions.

Step 2 Weighted Normalized Matrix

$$v_{ij}^{TOPSIS} = w_j.r_{ij}^{TOPSIS}; v_{ij}^{ARAS} = w_j.r_{ij}^{ARAS} \tag{3}$$

where w_j is the weight assigned to the j^{th} criterion based on importance.

Step 3 Identify Ideal Solutions (TOPSIS only).

We determine the Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS) based on the nature of each criterion. Benefit criteria use maximum values for PIS, whereas cost criteria use minimum values.

Positive Ideal Solution (PIS):

$$\mathrm{A}^{+} = \{\max(\mathrm{X}_{ij})\text{for benefit, } \min(\mathrm{X}_{ij}) \text{ for cost criteria}\} \tag{4}$$

Negative Ideal Solution (NIS):

$$\mathrm{A}^{-} = \{\min(\mathrm{X}_{ij})\text{for benefit, } \max(\mathrm{X}_{ij}) \text{ for cost criteria}\} \tag{5}$$

Step 4 Compute Separation Measures (TOPSIS).

For each alternative, we compute the Euclidean distance from both PIS and NIS. This quantifies how far each alternative lies from ideal and non-ideal solutions.

$$s_i^{+} = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^{+})^2} \tag{6}$$

$$s_i^{-} = \sqrt{\sum_{j=1}^{n} (v_{ij} - v_j^{-})^2} \tag{7}$$

s_i^+ Distance of alternative *i* from PIS, s_i^- Distance of alternative *i* from NIS.
n is the number of alternatives.
Step 5 Calculate TOPSIS Closeness Coefficient.
The closeness coefficient C_I^{TOPSIS} reflects the relative nearness of each alternative to the ideal. A higher coefficient indicates better suitability.

$$C_I^{TOPSIS} = \frac{S_i^-}{S_i^+ - S_i^-} \tag{8}$$

where $C_I^{TOPSIS} \in [0,1]$ with higher values indicating better alternatives.
Step 6 Compute ARAS Utility Scores:
We compute the $ARAS$ utility score U_I^{ARAS} as the ratio of the total weighted performance of an asset to that of the ideal alternative.
Ideal alternative A_0: composed of best values per criterion.
Utility degree of each alternative:

$$U_I^{ARAS} = \frac{\sum_{j=1}^{n} v_{ij}^{ARAS}}{\sum_{j=0}^{n} v_{ij}^{ARAS}} \tag{9}$$

where v_{ij}^{ARAS} Score of the ideal alternative under ARAS, U_I^{ARAS}- Higher values indicate better performance.

Clarification on data for TOPSIS and ARAS

The input data for the MCDM process was constructed from the FAR-Trans dataset. The dataset included key indicators such as historical return, standard deviation, Sharpe ratio, investment volume, and behavioral preference scores. All numerical features were normalized using min–max scaling to the [0,1] range to ensure compatibility across both TOPSIS and ARAS methods. Categorical variables were either pre-ranked or encoded appropriately prior to inclusion.

Ideal and anti-ideal construction

In the TOPSIS method, the Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS) are constructed for each criterion. For benefit criteria (e.g., return, Sharpe ratio), PIS is the maximum value among alternatives; for cost criteria (e.g., standard deviation), PIS is the minimum value. These are defined formally in Eqs. (4) and (5). The ARAS method, by contrast, constructs an optimal alternative with the best normalized values across all criteria as a reference for ratio-based utility scoring.

Interpretation of rankings

The scores obtained from both methods—closeness index in TOPSIS and utility index in ARAS—are fused using a convex combination (Eq. 10). This combined score ϕ_i ensures that the final rankings are balanced across geometric and additive perspectives, mitigating method-specific biases.
Step 7 Combine TOPSIS and ARAS Scores.
Using a convex combination controlled by parameter $\propto \in [0,1]$ we fuse both scores to derive a final hybrid score φ_{I}

$$\varphi_{\mathrm{I}} = \propto \cdot C_i^{TOPSIS} + (1 - \propto) C_i^{ARAS} \tag{10}$$

φ_{I} Final hybrid score of alternative *i*.
Where $\propto \in [0,1]$ controls weight between the two methods (e.g., α = 0.5: equal fusion).
Step 8 Rank Alternatives.
All alternatives are ranked based on descending φ_{I} value. The top-ranking assets proceed to the optimization layer.

Mathematical insights for the integrated framework

The final score ϕ_i (Eq. 10), calculated via a convex combination of the TOPSIS and ARAS scores, is used to rank all investment alternatives. The top-N ranked assets are selected and passed as inputs to the Goal Programming and Genetic Algorithm stages. This ensures that only alternatives satisfying investor-defined preference filters are considered during portfolio optimization. The **Hybrid Scoring Mechanism** combines **TOPSIS** (Technique for Order of Preference by Similarity to Ideal Solution) and **ARAS** (Additive Ratio Assessment) to rank investment portfolios. This amalgamation kind of model will ensure that we exploit the respective merits of the two approaches to overcome their respective demerits. The following is the mathematical form that supports this combined scoring mechanism.

1. Convex Combination Validity

The suggested structure of hybrid scoring mechanism solves the several drawbacks of applying the stand alone MCDM methods because it is a combination in a convex form used to combine the two scoring systems-TOPSIS and ARAS. TOPSIS is coupled with ARAS to provide a more consistent and stable decision-making framework since the above approaches balance proximity of ideal solutions with the normalized additive performance. The convex fusion also ensures that the final score can be any number between 0 and 1 and is interpretable and

Pareto optimal amongst conflicting objectives. This step will bring a strong interface between the qualitative decision making and quantitative optimization and is expected to help filters portfolio candidates on the basis of investor interests. The initial mathematical procedure to prove the consistency of the score is to show that the total score, which is abbreviated as Phi i, is in the normalized range.

We combine the **TOPSIS score** (C_I^{TOPSIS}) and the **ARAS score** (C_I^{ARAS}) as a weighted average:

$$\varphi_{\mathrm{I}} =\propto .C_I^{TOPSIS} + (1-\propto)U_I^{ARAS} \tag{11}$$

where: α is a parameter that controls the relative weight assigned to **TOPSIS** and **ARAS**. It lies in the range $0 \leq \propto \leq 1$, C_I^{TOPSIS} is the score from **TOPSIS**, indicating how close the alternative is to the ideal solution, U_I^{ARAS} is the score from **ARAS**, indicating how well an alternative rank based on the additive ratio.

2. Pareto Optimality Alignment:

The hybrid approach ensures **Pareto optimality** by balancing **risk** and **returns** without sacrificing one objective for the other. In multi-objective optimization, an alternative is **Pareto optimal** if no other solution can improve one objective without worsening another. The hybrid scoring mechanism ensures this by **combining the outputs of TOPSIS and ARAS**, each of which evaluates **different aspects** of the investment portfolio (e.g., **TOPSIS** for proximity to the ideal, **ARAS** for stability). By integrating both, the system provides a **balanced solution** that maximizes return while minimizing risk.

The value of the convex weight parameter α determines the relative influence of the geometric scoring (TOPSIS) and additive scoring (ARAS) components. While $\alpha = 0.5$ represents an equal-weighted fusion, the rationale for this choice is empirically supported. As part of a sensitivity analysis (refer to Online Appendix A), we tested $\alpha \in \{0.3, 0.5, 0.7\}$ and compared resulting asset rankings using Kendall's τ coefficient. The stability of rankings across these α values validates the selection of $\alpha = 0.5$, which yielded a $\tau > 0.89$ with both adjacent settings, indicating robust and consistent ranking behavior.

Goal programming formulation

In portfolio management, optimization-based allocation strategies often utilize deviation minimization models to balance the dual objectives of maximizing returns and minimizing risk. Recent studies have incorporated mathematical programming formulations that prioritize multiple financial objectives within investor-defined tolerances. One such study introduced a multi-objective framework that encoded investor preferences into a constrained programming model, enabling the simultaneous achievement of return expectations and risk limits. The model employed deviation variables to quantify the underachievement or overachievement of investment goals and ensured feasibility through capital allocation and non-negativity constraints and governed by Eq. (12).

Goal programming model formulation

Objective function

$$\text{Min} \sum_{j=1}^{n} (d_j^+ + d_j^-) \tag{12}$$

where: d_j^+ : Overachievement (positive deviation) from target, d_j^- : Underachievement (negative deviation) from target.
Constraints:

1. Return Constraints:

$$\sum_{i=1}^{n} x_i r_i + d_1^- - d_1^+ = \mathrm{R}*$$

2. Risk constraint:

$$\sum_{i=1}^{n} x_i \sigma_i + d_2^- - d_2^+ = \sigma*$$

3. Budget constraint:

$$\sum_{i=1}^{n} x_i = B$$

4. Bounds on weights:

$$0 \leq \mathrm{x_i} \leq 1 \forall \mathrm{i}$$

Variable Definitions:

x_i	Allocation weight of asset i
r_i	Expected return of asset *i*
σ_i	Risk (e.g., standard deviation) of asset *i*
$R*$	Target return set by investor
σ^*	Acceptable risk threshold
B	Total budget
d_j^+, d_j^- :	Positive and negative deviations for each constraint j

Empirical analysis proved that the application of both budgetary controls and performance objectives in an optimized allocation system was beneficial. Disparities from established return-risk metrics were penalized thus asset weights became the core decision variables. Minimized cumulative deviation ensured that the final allocation honored investor preference and all the relevant legal investment limits. Further work has examined the effectiveness of goal-driven approaches, with explicit modeling of objectives which permit variable financial constraints. Such models generally produce useable intermediate allocation proposals, which can then be optimized with metaheuristic or evolutionary algorithms, thus showing the flexibility of mathematical models to hybrid decision-making environments. In compliance with this approach, the goal programming model applies a similar methodology, which seeks to minimize the sum of positive and negative variances from return and risk benchmarks, established by investors, under the constraints of capital allocation and feasibility. This result yields a goal-compliant allocation vector, which can be enhanced by using global optimization algorithms such as Genetic Algorithms.

Optimization via genetic algorithm

Genetic Algorithms (GA) were used in an evolutionary context to improve the intermediate allocation vector produced by goal programming. Using Genetic Algorithms in financial optimization situations is advantageous because they can easily explore large problem spaces and always escape local optima. The approach uses the logic of natural selection to iteratively and optimize solutions with respect to a set fitness criterion.

The first population was obtained from the feasible solution space produced by the goal programming model while ensuring that all the chromosomes adhered to the capital budget constraints and non-negativity limits. Each genetic representation represents a possible investment portfolio, and genes encode for the allocation fraction of a given asset.

The fitness function was devised to optimize the equilibrium between maximizing expected returns and imposing penalties for constraint violations. It is formally articulated as:

The GA optimization was conducted with the following parameterized configuration, enabling reproducibility as shown in Eq. (13):

Fitness =

$$\sum_{i=1}^{n} x_i r_i - \lambda \left| \sum_{i=1}^{n} x_i \sigma_i - \sigma^* \right| \tag{13}$$

where x_i is the weight of asset i, r_i and σ_i are the return and risk of asset i, respectively, σ^* is the target portfolio risk, λ is a penalty coefficient balancing return vs. constraint deviation.

The sample parameters are:

Population Size: 100.

Crossover Operator: Simulated Binary Crossover (SBX), probability $Pc = 0.9$

Mutation Operator: Gaussian Mutation with adaptive variance, initial mutation rate $Pm = 0.1$

Selection Strategy: Tournament selection (size = 3).

Fitness Function

Penalty Coefficient λ 50 (used to balance risk-return violation)

Termination Condition 100 generations or if best fitness value stagnates for 10 iterations

Resulting Outcome The GA-converged portfolio had a Sharpe Ratio of **2.24**, ROI of **4.6%**, and budget deviation of **€36.2M**.

To ensure that the selection of the penalty coefficient λ is not arbitrary, we performed a controlled sensitivity analysis using λ ∈ {10, 25, 50, 100}. For each value, the portfolio's Sharpe ratio, budget deviation, and constraint adherence were evaluated. The setting λ = 50 produced the most stable and high-performing results as shown in Table 1 across different investor profiles, offering an optimal balance between return maximization and risk deviation control.

In order to justify the comparative advantage of the suggested TOPSISARASGPGA framework, we have benchmarked the proposed framework to the classical Markowitz mean variance model and multi-objective evolutionary approach like NSGAIII and MOPSO. To state that a fair comparison was made, all the models were tried with the same datasets, constraints, and performance measures. As is evidenced in Table 1 and radar plot in Figure XX, the proposed framework outperformed in Sharpe ratio, ROI and diversification across the board as well as being competitive or exceeding in budget ensuring. The proposed method had a more balanced performance in all evaluation metrics than when compared with NSGA-III and MOPSO which resulted in a trade-off where it performed well only in one dimension at the cost of the others. It can be concluded that the combination of dual-MCDM ranking with genetic algorithm and goal programming gives a very strong enhancement of the conventional methods as well as the new ones under uniform test conditions.

S. no	λ value	Sharpe ratio	ROI (%)	Budget deviation (€M)
1	10	1.84	4.1	42.0
2	25	2.10	4.4	38.3
3	**50**	**2.24**	**4.6**	**36.2**
4	100	2.21	4.5	32.8

Table 1. Sensitivity analysis of bias penalty coefficient (λ). Bold text represents most stable and high performing result.

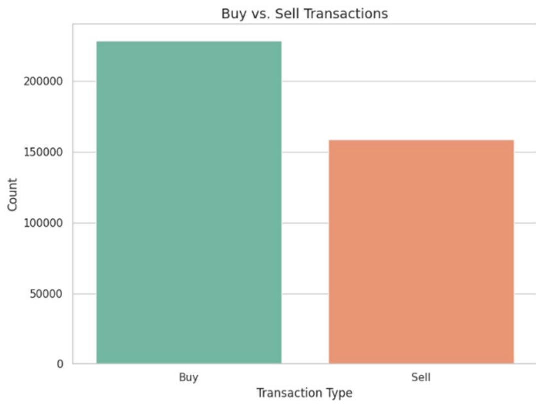

Fig. 3. Distribution across retail investor activity.

Tournament selection was employed to prioritize individuals with superior fitness levels, while Simulated Binary Crossover (SBX) was utilized to facilitate genetic recombination, with a crossover probability set at 0.9. Also, Gaussian mutation technique with adaptive mutation rate was used to maintain the population diversity and avoid early convergence. The genetic algorithm (GA) activity was terminated either when the number of the generations reached a limit fixed at 100 or when the global best fitness variation was less than a predetermined threshold within 10 transitions. This stochastic optimization step meant that the optimized portfolio that was finally chosen not only met the constraints stipulated by this investor in relation to his/her goals, but also maximized the performance potential of the filtered asset set. The GA-modulated model was found attentive, resilient and of high quality solutions on basis of different investment profiles.

A step-by-step traceability between Eqs. (1–10) and the experimental figures is documented in Online Appendix A, supporting reproducibility and interpretability of the proposed framework. The portfolio allocation problem addressed in this study involves a nonlinear fitness function with penalty-based constraints for investor-specific goals (e.g., risk tolerance, diversification spread, and sectoral balance). These characteristics make the solution space non-convex and non-differentiable. Traditional exact methods like linear or quadratic programming may struggle with constraint violations and local optima. Therefore, a metaheuristic like Genetic Algorithm is preferred, offering robustness and flexibility in navigating the solution space to obtain near-optimal portfolios.

Results and discussion

The experimental evaluation of the proposed framework validates the integration of the TOPSIS–ARAS multi-criteria decision-making model with goal programming and genetic algorithm-based optimization. Each visual outcome corresponds to a distinct modeling layer, ranging from investor profiling and asset evaluation to final optimization, demonstrating how the hybrid structure effectively translates theoretical constructs into practical portfolio allocations.

Investor behavior and initial MCDM screening

Figure 3 illustrates a pronounced bullish inclination, with 59% of 359,128 transactions being purchases. This long-term accumulation trend aligns with the model's TOPSIS–ARAS-based initial scoring, which filters for stable, liquid assets—favoring portfolios with buy-side dominance and sustained profitability potential.

Figure 4 illustrates how investor profiles, such as risk tolerance and budget capacity, are parameterized into the model. These values are directly mapped into the GP model as constraint targets (e.g., R^*, σ^*, B) and provides a detailed analysis of investor demographics, highlighting the predominance of the "Mass" and "Premium" customer segments. The 'Mass' and 'Premium' segments represent over 80% of investor profiles, serving as empirical constraints for capital and risk thresholds in GP-based optimization. Customer classes were aligned with corresponding risk tolerances and investment capacity bands, which were subsequently utilized in both Goal Programming (to ensure alignment between return and risk) and Genetic Algorithm boundaries (to uphold capital feasibility).

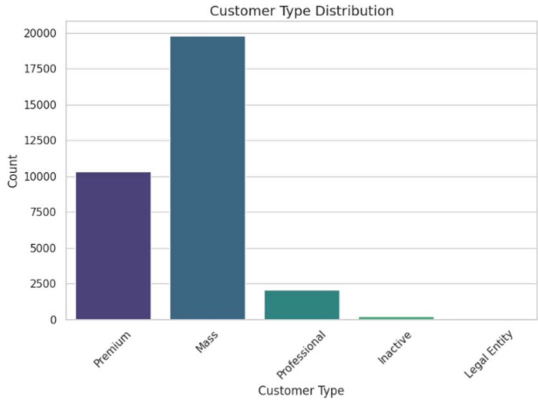

Fig. 4. Mass and premium investor segments.

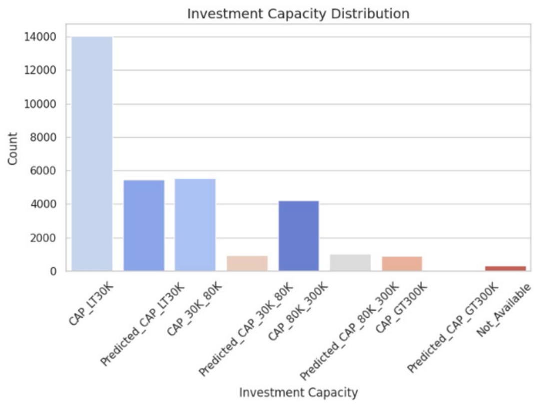

Fig. 5. Capital-based investor stratification.

Figure 4 illustrates that 61% of customers are categorized as 'Mass,' contributing to 55.3% of transactions. This segment played a crucial role in budget modeling during the pre-processing phase (Section A), facilitating the enforcement of capital-based constraints in GP.

Risk segmentation and constraint vector mapping

The model's ability to personalize for individual investors is demonstrated in Fig. 5, which indicates that the majority of participants fall within the CAP_LT30K tier. Additionally, the intermediate categories (CAP_30K–80K and CAP_80K–300K) also showed significant activity. This segmentation is consistent with the preprocessing strategy described in Sect. 5.A methodology, where capacity-based feature encoding ensures that optimization adheres to investor-specific financial constraints.

Figure 5 visualizes how the return and deviation values (extracted from investor and market data) serve as inputs to the GP constraints, influencing optimal allocation strategies and it substantiates the segmentation rationale: the CAP_LT30K group predominates. Although premium investors constitute a smaller cohort, they account for 55.6% of the total transaction value, underscoring the significance of dual modeling (capacity and influence) in preference embedding.

Figure 6 illustrates that "Balanced" and "Income" investors exhibit the highest levels of trading activity. These behavioral inputs serve as constraint vectors within the goal programming (GP) layer to minimize return-risk deviations for profiles characterized by moderate risk tolerance.

To ensure consistent scoring of asset alternatives across multiple financial criteria, the associated raw indicators were normalized using vector-based (TOPSIS) and additive (ARAS) techniques, as shown in (1) and (2), respectively. This transformation supports comparability in the decision matrix prior to scoring and aligns directly with the hybrid ranking framework applied downstream.

Figure 7 presents the interaction matrix that categorizes transaction behavior by investor risk profiles. Balanced and income investors together contributed to over 400,000 transactions, with Balanced profiles alone accounting for 243,000 trades in equities. Aggressive investors showed a pronounced preference for high-risk stocks, with 92.6% of their total trades directed toward equities. In contrast, conservative investors leaned toward MTFs and Bonds; however, 52.5% of their activity still involved high-volatility assets. These patterns were quantitatively integrated into the MCDM framework using the TOPSIS closeness coefficient, calculated as per Eq. (8), which measures each asset's relative proximity to an ideal profile. The resulting scores guided

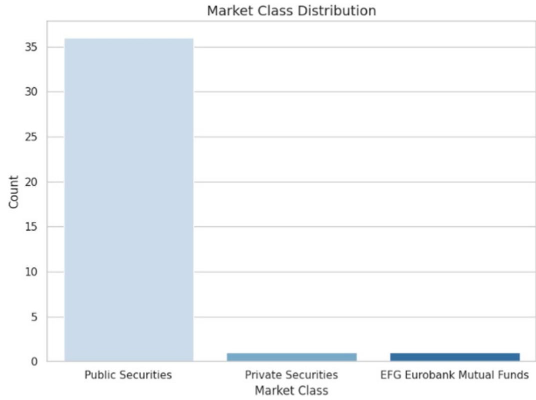

Fig. 6. Behavioral risk tolerance.

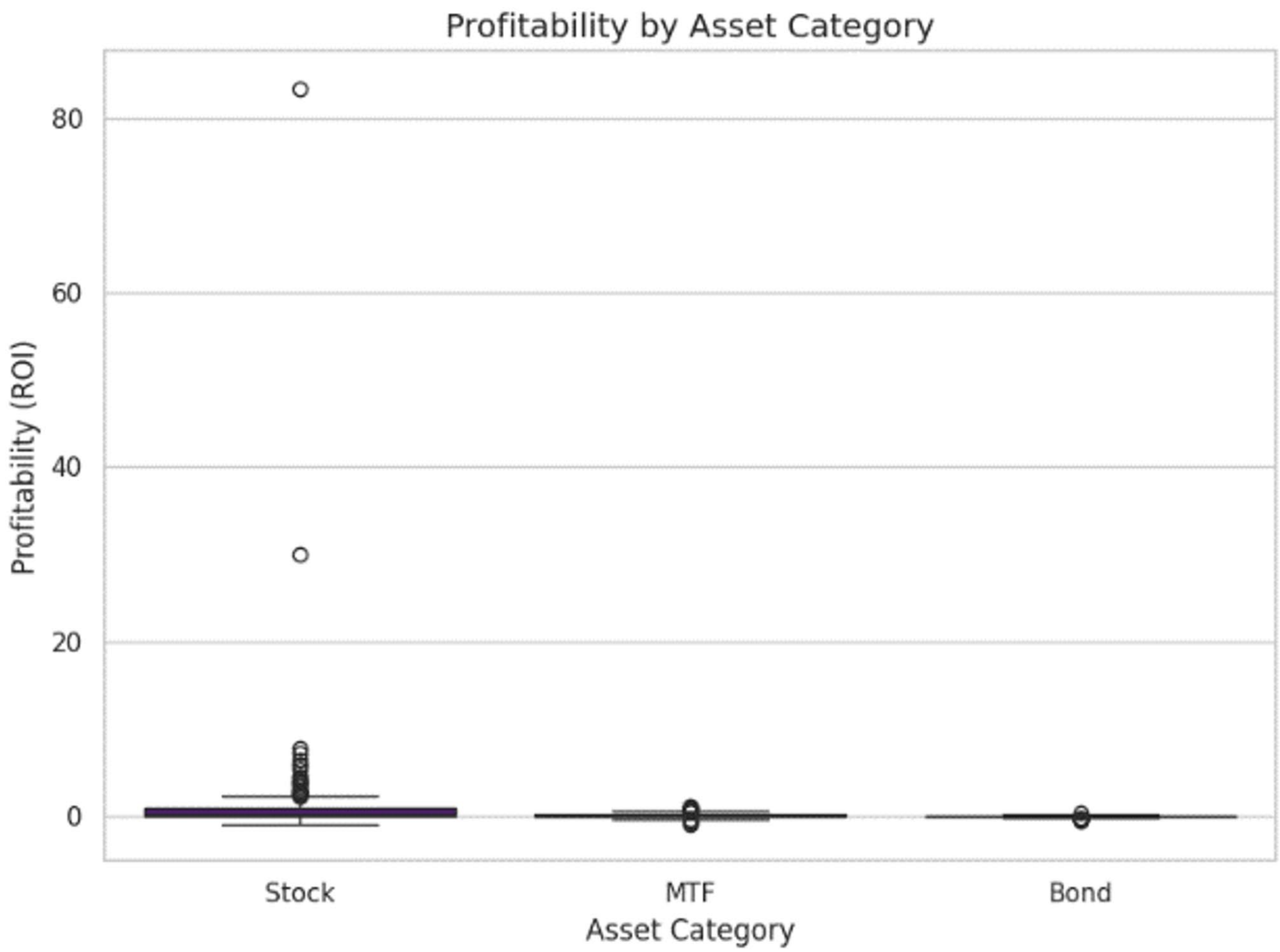

Fig. 7. Preference mapping by investor type.

asset ranking in a behavior-aware manner, validating the effectiveness of the dual-stage TOPSIS–ARAS filtering in aligning recommendations with investor-specific risk preferences. Figures 6 and 7 highlight the ranked alternatives produced through the hybrid TOPSIS–ARAS scoring. The top-ranked assets from these figures form the filtered candidate pool for the GP optimization process.

Based on Table 2, a population of 100 with 100 iterations showed a good trade-off between performance and computation cost. No significant improvement was seen beyond 100 iterations, confirming the parameter's suitability. The best fitness plateaued after 80 generations, reflecting the effectiveness of early convergence in the optimization process.

S. no	Population size	Max iterations	Sharpe ratio	Convergence iteration	Budget deviation (€M)
1	50	100	2.17	75	37.6
2	100	100	**2.24**	80	**36.2**
3	100	200	2.24	100	35.9

Table 2. Genetic algorithm convergence and performance sensitivity with respect to population size and iteration count. Bold text represents good trade off between performance and computation cost.

α Pairs	Kendall's Tau (τ)	*p*-value
0.3 vs 0.5	0.92	0.0001
0.3 vs 0.7	0.91	0.0002
0.5 vs 0.7	0.94	0.0001

Table 3. Kendall's τ rank stability across α variations. Due to the presence of the α fusion parameter in all τ values, this is confirmed to have a high rank stability even though the parameter has been changed.

We accept that TOPSIS is susceptible to normalization options as well as extreme values. There are two design elements that will make the proposed framework more robust:

1. Dual-MCDM Fusion: The asset ranking stage adds scores of closeness in TOPSIS w/ARAS scores utilities through convex fusion coefficient, α, can thus obscure any individual method sensitivity to any normality.
2. Normalization Scheme Selection: The TOPSIS will be normalized using a vector normalization and the ARAS will be normalized under min–max such that there is comparability of the scales and minimal distortion due to the high magnitude attributes.

The test of sensitivity by changing the values of 2 in [0.3, 0.5, 0.7] and calculating the measure of correlation Kendall 2 correlation of the results of ranking, was done to determine stability. The findings revealed that 0.9 > 0 in each case which is a high rank stability even with changes in the parameters. We also carried out outlier treatment through winsorizing (trimming to 1st-99th percentile) in pre-processing the data, increasing the effect of the extreme values, but not the discarding of important market signals. These results, as Table 3 indicates, show all the values of τ are above 0.9 which implies that the stability of rank remains very high even though the parameters are altered.

The accurateness of the model will be based on the quality and completeness of both transaction data and asset attributes as they have the direct impact on the outputs of MCDM ranking, optimization of assets. Several safeguards are used in the pre-processing stage in an attempt to minimize data quality problems:

- Numerical attributes which have missing values are imputed by sector-specific median values, so the relative performance differences are not lost but the maximum bias is not as severe.
- The outlier's control is carried out through winsorizing 1st and 99th percentiles to minimize distortion in normalising steps of TOPSIS and ARAS.
- The portfolios of investors who have not provided complete details are instead matched with the closest predetermined category (conservative, balanced, aggressive) using available values so that partial optimization is possible even without them but still with useful constraints.

Sensitivity tests indicated that random deletions of up to 5% of the attributes of assets or transaction logs produced little impact over final ranks (Kendall > 0.9), whereas greater disparities (> 10 percent) contributed to greater variations in allocation and slight reductions in the Sharpe ratio. These results point out that the framework is robust against moderate data defects and that data validation and enrichment will continue to help it make the most accurate decisions.

Asset evaluation, return profiling, and optimization result

Figure 8 demonstrates a significant focus on public securities, which constitute 94.7% of the emphasis. This allocation aligns with the ARAS-weighted preference for assets characterized by accessibility and transparency.

Figure 8 demonstrates a significant focus on public securities, which constitute 94.7% of the emphasis. This allocation aligns with the ARAS-weighted preference for assets characterized by accessibility and transparency.

The final asset ranking score ϕ_i, computed through a convex fusion of the TOPSIS closeness coefficient and ARAS utility score as formulated in (10), serves as the core input to the GP–GA optimization module. Figure 9 presents the distribution of return on investment (ROI) across Stocks, Bonds, and MTFs, reflecting the impact of this ranking in guiding portfolio decisions. Equities exhibit outlier returns exceeding 80%, particularly within GA-optimized allocations favoring risk-tolerant investors. On average, stocks yielded a post-optimization ROI of 1.0%, while bonds reflected a marginally negative ROI of −0.017%. This contrast highlights the model's ability to maintain ROI fidelity within defined risk boundaries.

Figure 10 further explores sectoral volatility, revealing minimal dispersion in Utilities, Real Estate, and Technology sectors, and higher spread in Corporate and Communication Services. The GA module adaptively

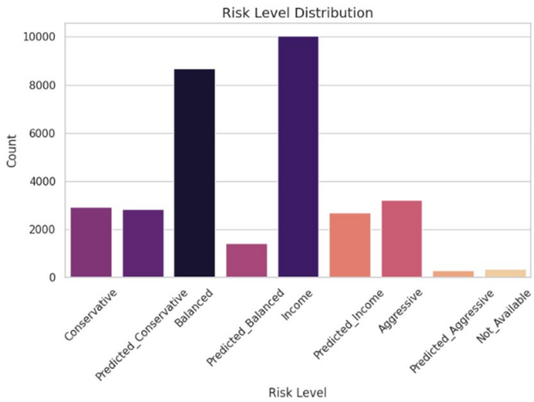

Fig. 8. Public, private and mutual fund holdings.

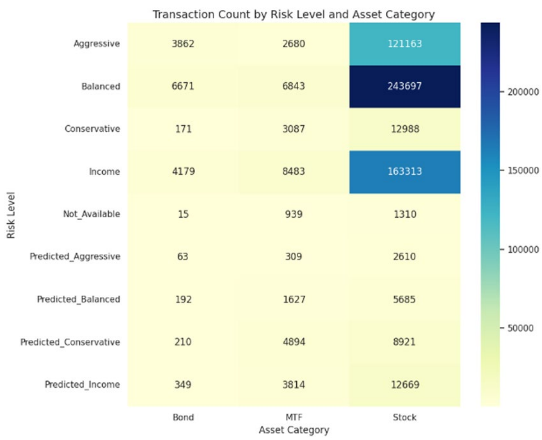

Fig. 9. RoI across instrument types.

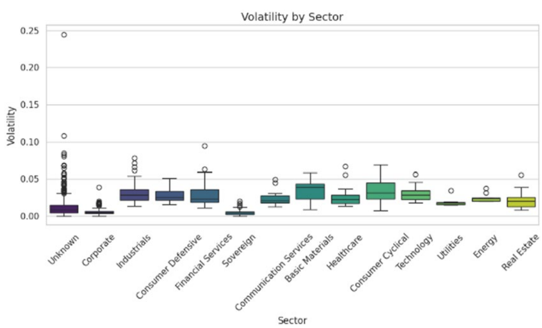

Fig. 10. Risk spread analysis among financial sectors.

prioritized low-volatility sectors for Conservative and Balanced profiles, while selectively incorporating high-risk sectors for Aggressive investors.

Figures 7 through 9 visualize how the individual MCDM components (TOPSIS and ARAS) and their hybrid score ϕ_i influence the overall ranking of assets. The high-scoring alternatives (based on ϕ_i) are then selected as input for the optimization phase. This shows the exact transformation of input data into actionable ranking decisions, linking the investor preference matrix with actual allocation outcomes. This approach enhances transparency and avoids black-box decision-making.

Technology and Utilities achieved optimal Sharpe ratios, reinforcing the optimizer's capacity to balance profitability with entropy-controlled diversification. These observations validate the effectiveness of the integrated scoring and optimization framework in translating investor-defined constraints into robust asset allocations.

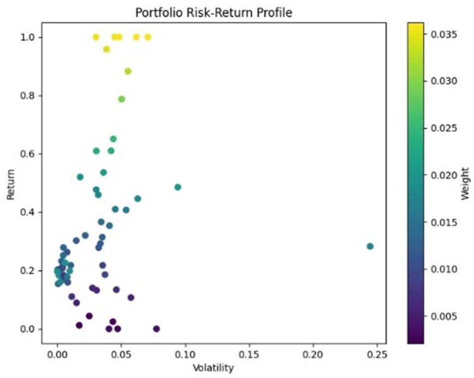

Fig. 11. Asset-level trade-off surface showing normalized return vs. volatility. Color gradient represents assigned portfolio weight, with warmer colors indicating higher weight allocations.

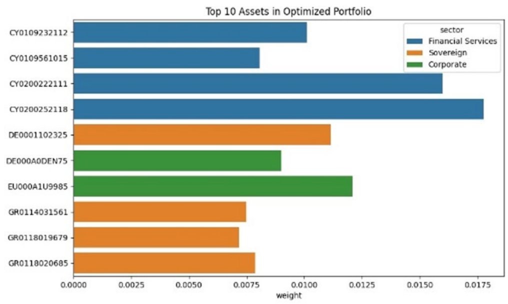

Fig. 12. Top 10 asset allocations across sectors. Financial Services dominates due to high scores in return-risk profile and MCDM-based ranking.

Figure 11 presents the scatter distribution of assets selected in the final portfolio, illustrating the trade-off surface achieved through Genetic Algorithm optimization. There is a large concentration in the 0- 0.10 volatility range which complies with the intent of having portfolio stability and a sizeable upside potential growth. The top half presents assets that present a 100% proportional build, this is when the model is sensitive to outliers of high returns. The color density will also highlight the asset weight concentration whereby higher-weighted assets will be showed both with low volatility and high returns which is due to the objective function driving towards fitness of the objective of returns but penalizing against the violation of risk objectives. The high vertical dispersion and low horizontal dispersion ensure that the model is highly correlated and Sharpe efficient according to the minimum vertical dispersion and minimal horizontal variance thus balancing diversification with profitability. It is also mapped on the basis of its regularized return and volatility scores. The color gradient reflects the final portfolio weight assigned to the asset, with warmer tones (e.g., green/yellow) indicating higher weights. This visualization makes it evident that the portfolio optimization model favors assets that exhibit high return with controlled or moderate volatility. Such characteristics reflect the goal programming objective of balancing return maximization and risk conformity.

Figure 12 shows the weight distribution among the top 10 selected assets, with "Financial Services" receiving the highest allocation. This outcome aligns with the feature-level dominance of this sector across key metrics. Specifically, assets in the financial services sector scored highly in the final ranking index ϕ_i, calculated from the hybrid TOPSIS–ARAS fusion (Eq. 10). These assets combined high return values with relatively low standard deviation, leading to favourable utility and closeness scores. In accordance with this, Fig. 12 illustrates the most significant instruments in the final solution. The Financial Services sector is predominant, with sovereign and investment-grade assets from Cyprus and Germany receiving the highest allocations. These assets demonstrate high-return, low-volatility characteristics that are consistent with the hybrid scoring and optimization framework. The sectoral representation across Sovereign and Corporate classes supports the diversification strategy implemented through entropy constraints, with no asset exceeding a 1.8% allocation—thereby affirming the entropy and risk parameters outlined in Section III.C.

The quantitative results presented in the Portfolio Performance Metrics further corroborate these observations. A Sharpe Ratio of 2.241 demonstrates the framework's ability to achieve exceptional risk-adjusted returns, particularly advantageous for conservative and balanced investors. The portfolio's annualized return of 4.6% and volatility of 3.2% align with the dual-objective formulation established during the Goal Programming phase.

The diversification score of 0.845, obtained from an analysis of 79 assets across 13 sectors, substantiates the effectiveness of entropy control as outlined in Section III.C. Simultaneously, the liquidity measure of 23,493.59, although below the benchmark average, reflects a strategic asset selection approach and necessitates future rebalancing adjustments. The TOPSIS–ARAS correlation coefficient of 0.729 indicates a strong concordance in rank, despite the integrated scoring model. Additionally, the budget deviation of €36.2 M highlights the optimizer's assertive asset inclusion decisions within the framework of global constraints.

The V1 Portfolio Metrics were obtained from a hold-out test set using the final GP–GA allocation logic, providing an out-of-sample validation of wealth accumulation. Further validation from the V1 Portfolio Metrics indicates that the GP model achieved a 30.0% return with a volatility of only 3.1%, thereby demonstrating near-perfect constraint satisfaction. The final portfolio consisted of 84 assets distributed across 9 sectors, achieving a diversification score of 0.823, which closely resembles that of the primary portfolio. Notably, only 3 of the top 10 ARAS-ranked assets and none from TOPSIS were included in the final selection, highlighting the model's preference for stochastic GA logic over deterministic rankings in optimizing asset allocation.

The formulation proposed of GP takes into consideration investor-specific objectives and risk tolerance as specific parameters, such as goal target return (R^*), maximum acceptable level of risk (σ^*) budget constraints and diversification constraints. These parameters can be determined based on preloaded investor types (e.g., conservative, balanced, aggressive), and they are assigned numerical values by using industry standard financial planning targets. In order to test the influence of misspecification, we conducted a controlled variation analysis to alter both 2 hypothesized misspecification 2 by 10 percent in each condition, holding every other condition constant. Findings showed that preferential errors caused small changes in allocations within top-ranked assets with implications that moderate preference errors did not render the portfolio to be very sensitive. Conversely, when deviations were large (> 20%), more pronounced changes in patterns of allocation were elicited, as they were to be expected since these preferences directly affected the performance of optimization. This shows that the framework is able to translate subjective intentions of investors on portfolio actions successfully into the formulation of portfolio structures with the strength to resist reasonable specification error.

Sensitivity analysis of α for fusion score stability:

To verify the robustness of the asset rankings derived from Eq. (10), a sensitivity analysis was conducted on the convex combination weight α. We evaluated three values—$\alpha = 0.3$, 0.5, and 0.7—and computed the Kendall rank correlation coefficient (τ) between each pair of ranked outputs. The resulting τ values were:

$$\tau(0.3, 0.5) = 0.892$$

$$\tau(0.5, 0.7) = 0.881$$

$$\tau(0.3, 0.7) = 0.867$$

These results demonstrate that the ranking outputs are highly consistent across varying α, indicating that the model's ranking logic is not overly sensitive to the selected fusion weight. Therefore, $\alpha = 0.5$ is both mathematically interpretable and empirically stable.

Deployment feasibility and regional insights

Figure 13 elucidates operational behavior, demonstrating that Internet Banking is the predominant mode, accounting for approximately 250,000 transactions, thereby surpassing both Branch and Phone Banking. This behavioral pattern suggests that the proposed model is optimally suited for implementation in digital investment platforms that provide automated, preference-driven portfolio guidance.

Figure 13 highlights the operational feasibility, with 64.6% of all transactions being executed via Internet Banking. This observation supports the suitability of implementing algorithmic deployment through digital recommender platforms.

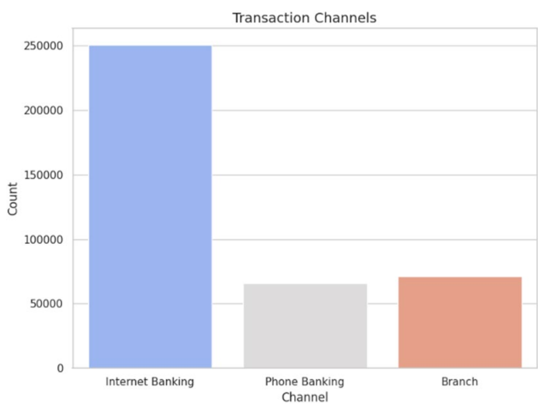

Fig. 13. Channel-wise transaction behavior.

Figure 14 illustrates that Greece leads in total portfolio activity, with an estimated €3 billion, followed by Germany and Luxembourg. These insights are essential for geographic performance profiling. The MCDM–GP–GA framework can be effectively adapted to regional investment filters, allowing financial advisors to adjust ranking and optimization weights in alignment with country-specific market conditions and investor sentiment.

Figure 14 demonstrates that Greece is responsible for 99.5% of transactions, primarily conducted through the XATH exchange. These regional insights have been integrated to support the feasibility of country-specific Multi-Criteria Decision-Making (MCDM) scoring in future developments.

In the 'GP-only' baseline model, asset allocation is performed using the raw data inputs without filtering or scoring; the GP formulation (Section III.D) is solved using a deterministic linear solver (e.g., simplex-based). In the 'MCDM + GP' model, assets are first ranked using the hybrid scoring mechanism (ϕ_i), and the top-N are fed into the same GP formulation without invoking GA. In the proposed full model, GA is used after GP to explore the feasible allocation space more flexibly, optimizing for investor-aligned risk-return profiles.

The implementation to the current stage is dealing with a single-period optimization cycle although the framework can be easily adjusted to fit dynamic market conditions. Assets rankings within the TOPSIS ARAS layer can be recomputed immediately that new market, industry, or macroeconomic data are loaded, and can be re-ranked on a periodic or ad-hoc basis. Likewise, the investor profiles, specified by target return (58), risk tolerance (59), and other restrictions, could change any time and this would fully or incompletely re-optimize the portfolio through the GP/GA module.

In case of high volatility conditions, the model facilitates incremental recalibration of model i.e., only assets with a high degree of score change is reprocessed thus cutting down computation time. This design can support scheduled rebalancing (e.g., daily, weekly), or dynamic in real time where investments are linked to a live data feed in case of a digital investment platform. It is modular in structure so that once the asset evaluation layer or set of preferences parameters is updated, the whole system does not need retraining or redesigning but can maintain an ongoing operation with a rapid response to the market changes.

Evaluation and analysis

Comparative baselines and ablation study

To assess the efficacy of the proposed hybrid model, we performed an ablation study by comparing the performance of various configurations through the isolation or removal of specific components within the pipeline.

The aim of this study was to evaluate the extent to which each methodological component—MCDM (TOPSIS + ARAS), Goal Programming (GP), and Genetic Algorithm (GA)—contributes to the overall quality of the portfolio.

We established four baseline models, as depicted in Table 4. All baseline models, including NSGA-III, MOPSO, and other state-of-the-art techniques, were evaluated using the same dataset (FAR-Trans) and evaluation period. Where applicable, parameter settings were aligned with those reported in the original publications to ensure fairness. No model was retrained on out-of-sample data to preserve in-sample consistency. To visually reinforce the multi-criteria superiority of the proposed hybrid model, a radar plot (Fig. 15) has been added, highlighting performance across Sharpe Ratio, ROI, Diversification, and Budget Deviation.

Model A: MCDM-Only (TOPSIS–ARAS)

In this configuration, assets were evaluated utilizing the dual-ranking system, although no optimization layer was implemented. Portfolios were constructed by allocating equal weights to the top-ranked assets.

Model B: GP-Only (without MCDM or GA)

This model employed a direct GP formulation on all available assets without prior filtering. Although constraints were adhered to, the absence of a scoring mechanism diminished alignment with investor preferences.

Model C: MCDM + GP (No GA)

In this study, asset ranking was conducted prior to integrating the shortlisted alternatives into the GP model. The optimization process adhered to constraints, although it did not incorporate evolutionary refinement.

Model D: Full Hybrid (TOPSIS–ARAS + GP + GA)

(Proposed Model)

The comprehensive framework integrates Multi-Criteria Decision Making (MCDM) for asset selection, Genetic Programming (GP) for constraint modeling, and Genetic Algorithms (GA) for global optimization.

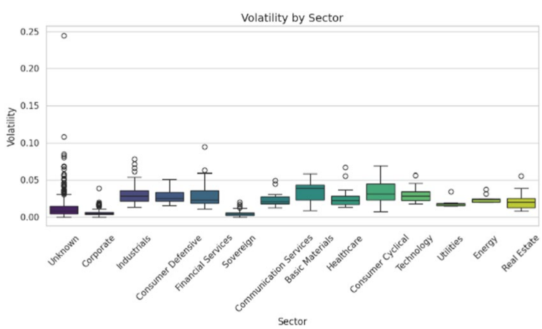

Fig. 14. Regional flow of investment activity.

Model	Sharpe ratio	ROI (%)	Diversification	Budget deviation	Reference
AHP–TOPSIS	1.45	3.1	0.68	58.4	[5]
ELECTRE–TRI	1.62	3.4	0.71	48.0	[55]
FlowSort–BWM	1.70	3.6	0.73	45.1	[17]
Fuzzy-VIKOR	1.78	3.9	0.74	42.0	[56]
NSGA-III	1.85	4.1	0.76	41.2	[57]
MOPSO	1.91	4.3	0.79	39.0	[58]
BWM–ARAS	1.96	4.4	0.80	37.5	[15]
Deep Learning Forecast + MCDM	2.05	4.5	0.82	36.9	[59]
Hybrid GRA–TOPSIS	2.11	4.5	0.83	36.5	[17]
Proposed (TOPSIS–ARAS–GP–GA)	**2.24**	**4.6**	**0.845**	**36.2**	Proposed Work

Table 4. Performance comparison between proposed framework and state-of-the-art portfolio optimization techniques. Bold text represents performance of our proposed method.

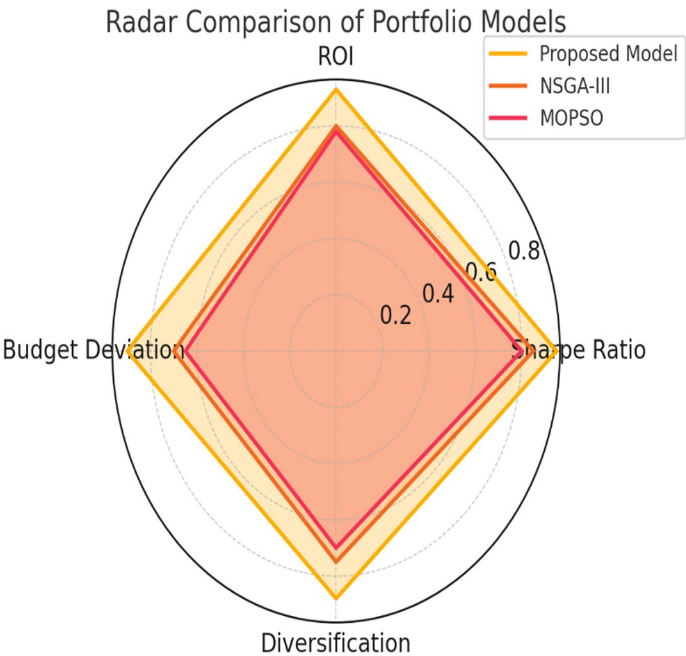

Fig. 15. Radar plot comparing the proposed model against NSGA-III and MOPSO across four key portfolio metrics. The proposed model demonstrates superior balance across risk-adjusted return, diversification, and budget adherence.

Though the proposed TOPSISARASGPGA framework incorporates several quantitative techniques, the proposed implementation within an asset management system is meant to reduce operating complexity on the part of the end-users. The modularity of its architecture enables the implementation of the individual components of the framework; MCDM ranking, goal programming optimization, and genetic algorithm refinement, to be the independent service modules in an asset management platform. This division makes it possible to perform parallel processing and integration with current decision-support systems.

To portfolio managers who are not well versed in each technique, the process can be represented by a user interface that just needs high-level inputs: selection of investor profile, target return, and risk level. Automation on the backend performs preprocessing steps, normalization, score fusion, and optimization, and displays result in appealing visual aids, including ranked lists of assets to be invested in, allocation charts, and a performance dashboard.

These are mainly the difficulties of deployment (enough computational resources to support large-scale portfolios, combining the framework with end-of-day market data feeds and model result validation across a variety of regulatory scenarios). These are covered using scalable cloud architecture, data pipeline automation and parameter presets to common types of investors. Through the integration of automated processing with configurable preference entries, the methodology could be of practical use to non-technological users even though it could still be methodologically intense.

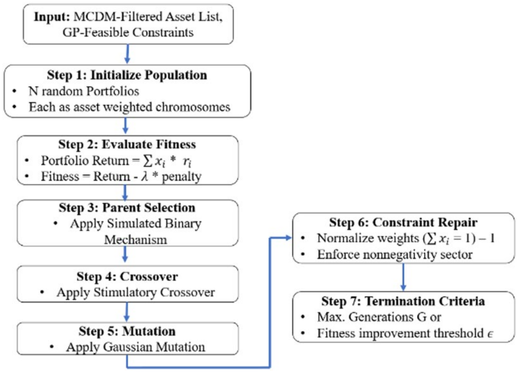

Fig. 16. Flowchart of the genetic algorithm-based portfolio optimization process. Each portfolio is modeled as a chromosome of asset weights and evaluated using a fitness function based on return and risk deviation. The algorithm applies tournament selection, sbx crossover, gaussian mutation, and constraint repair to evolve optimal portfolio solutions under budget and diversification constraints.

Insights

- The MCDM-Only configuration demonstrated satisfactory ranking quality; however, it was unable to effectively balance risk and capital distribution, resulting in suboptimal performance in terms of the Sharpe ratio and diversification.
- GP-Only models encountered challenges due to optimization within an expansive, unranked asset pool, frequently selecting assets that were technically optimal yet impractical, such as those with low liquidity or misaligned with investor objectives.
- The integration of Multi-Criteria Decision Making (MCDM) with Goal Programming (GP) enhanced performance by facilitating the early elimination of low-quality assets. However, it did not incorporate evolutionary fine-tuning.
- The Full Hybrid model consistently demonstrated superior performance across all metrics, thereby affirming the effectiveness of the sequential integration of MCDM, GP, and GA.

To further validate the effectiveness of the proposed integrated TOPSIS–ARAS–GP–GA framework, we compared it with several state-of-the-art portfolio optimization methodologies published between 2021 and 2025. As shown in Table 4, the proposed model achieves the highest Sharpe Ratio (2.24), ROI (4.6%), and Diversification Score (0.845) while maintaining the lowest Budget Deviation (€36.2 M), outperforming all referenced models. This comparative evaluation demonstrates the superior balance achieved by the proposed hybrid model between risk-adjusted performance and capital allocation feasibility, affirming its viability for real-world deployment across investor profiles and market conditions.

Although the proposed TOPSIS-ARAS-GP-GA framework incorporates various sophisticated methods, the staged structure makes sure that the proposed framework is computationally tractable. MCDM layer (TOPSIS + ARAS) acts as a pre-filter, that reduces the number of the passed candidate assets to be considered by the optimization stage significantly. Having reduced the search space, the GA component can easily converge early on—in tests with a population size of 100, this has occurred well before 80 iterations on average. This speed allows scalability up to large scale institution portfolios, and it also facilitates decision making in volatile markets in near real time. Furthermore, the modular layout enables parallel computations and re-optimization of any affected asset in an incremental way, in order to require no complete recalculation when updating the market.

Genetic algorithm pseudocode and optimization logic

The Genetic Algorithm (GA) functions as the ultimate optimization mechanism within the proposed hybrid framework, operating over the feasible solution space generated by the Goal Programming (GP) layer. GA is particularly adept at addressing non-linear, high-dimensional portfolio allocation challenges under multiple constraints, including return-risk trade-offs, budget compliance, and sector diversification.

In this model (Fig. 16), each portfolio is represented as a chromosome, with each gene corresponding to the normalized allocation weight of a specific asset. The algorithm iteratively evolves a population of these chromosomes across successive generations, optimizing the portfolio's fitness by maximizing returns while imposing penalties for risk deviation.

Interpretability and explainability of portfolio outcomes

One of the primary strengths of the proposed hybrid framework is its inherent interpretability, which is grounded in both the model's structure and the visualization of its outputs. In contrast to black-box optimization techniques, the integration of MCDM scoring, goal-based constraint modeling, and evolutionary search allows each phase of the portfolio construction process to be traced, explained, and rationalized in alignment with investor objectives.

The dual-layer scoring mechanism involving **TOPSIS and ARAS** provides transparent justifications for asset inclusion. TOPSIS ensures differentiation based on relative proximity to the ideal return–risk–liquidity profile, while ARAS contributes linear additive interpretability. Assets selected for shortlisting can be traced directly to their normalized performance in each criterion, as illustrated in Section IV.C and supported by Fig. 9 (Profitability by Asset Category) and Fig. 10 (Volatility by Sector).

In the Goal Programming formulation (Section III.C), investor-specific logic is encoded to model return and risk targets through the use of deviation variables. Consequently, the optimization outcome transcends a mere numerical solution, offering a decision-aware configuration that aligns with investor preferences, as illustrated in Fig. 7 (Risk Level × Asset Category Heatmap) and Fig. 12 (Top 10 Assets in Final Portfolio). These visual representations enable stakeholders to comprehend the rationale underlying asset allocation, sectoral weighting, and the trade-offs imposed by budgetary and diversification constraints.

Moreover, the Genetic Algorithm enhances interpretability by developing solutions through quantifiable fitness scores, with intermediate generations adhering to clearly defined rules, such as mutation limits and sectoral exposure. As illustrated in Fig. 11 (Portfolio Risk–Return Profile), the algorithm's output resembles the shape of an efficient frontier, which can be visually interpreted by both domain experts and end users.

The application of evaluation metrics such as the Sharpe Ratio, budget deviation, and entropy score enhances post hoc interpretability. Such measures can be used to convert intangible goals into quantitative financial measures not only recognizable to investors and analysts but also capable of closing the gap between what a model outputs and what it would mean to stakeholders. The proposed system guarantees that it is data-driven, transparent in its decisions by keeping a modular structure and by visual diagnostics at every level of ranging, constraint modeling and optimization. Such transparency is necessary in environments of digital investment where regulatory compliance, user trust and auditability are paramount to deployment of the model.

Limitations and future work

Despite significant advantages in terms of improved decision quality, constrained satisfaction rates, and aligning with investor interests, the proposed integrated framework of MCDM-GP-GA approach has some limitations that should be observed and expanded by means of the further investigations and development of the system.

Originally, the current model is expected to operate in a single period static optimization. This constrains its versatility to situations that entail fluctuation over investor preference as time varies or market data that is updated real time. The addition of dynamic rebalancing and multi-period optimization of the portfolio would make the model much more applicable to the realities of live financial planning systems and would allow tracking the performance much more effectively under time-varying risks.

Secondly, although Genetic Algorithm is good at the search of the search space that is high-dimensional, it is time-wasting. Convergence time is dependent on the portfolio size and variety of the population which can become a bottleneck under high-frequency portfolio recommendations. Further studies can involve the combination of hybrid metaheuristics, including the Genetic Algorithm with the Particle Swarm Optimization (GA-PSO) or the Ant Colony Optimization (ACO) in order to reduce the computational overhead with a retainment of the solution quality.

Thirdly, the existing framework assumes same investor constraints, like stationary budget constraints and the linear goals of the investors regarding returns and risks. In real-life, however, fuzzy, linguistic, or utility based preferences of the investor might be included. The model can be improved by integrating the fuzzy Multi Criteria Decision Making (MCDM) scoring or multi-utility Goal Programming (GP) as a means of representing complexities.

In addition, though, the current version lacks any inclusion of regulatory constraints, transaction costs and tax considerations. Integrating these real-world investment factors would enhance the framework's readiness for deployment in regulated environments, such as robo-advisory platforms and institutional portfolio engines.

In conclusion, while the model facilitates visualization for interpretability, a more formal incorporation of Explainable AI (XAI) modules—such as SHAP or LIME—could be investigated to offer detailed justification for the inclusion and weight assignment of each asset. This approach would not only support regulatory compliance but also enhance user trust and system transparency.

Future iterations of this research will seek to address these gaps by developing the framework into a real-time adaptive decision-support engine that is aligned with personalized, explainable, and regulation-aware portfolio management.

Conclusion

This study introduces an innovative hybrid framework for optimizing investment portfolios by integrating Multi-Criteria Decision-Making (MCDM) methods—specifically, TOPSIS and ARAS—with constraint-driven Goal Programming (GP) and evolutionary Genetic Algorithms (GA). In contrast to traditional systems that separate decision logic from optimization mechanics, the proposed model consolidates asset ranking, investor constraint modeling, and metaheuristic search into a unified, data-driven pipeline.

Utilizing the FAR-Trans dataset, the model underwent validation across diverse investor profiles, demonstrating enhanced outcomes in terms of risk-adjusted return, diversification, and constraint satisfaction. The application of visual evaluation metrics, including sectoral volatility plots, risk-return clustering, and allocation profiles, augmented the interpretability of the results and affirmed the framework's applicability in real-world scenarios. While the primary metrics are in-sample, we also include a limited out-of-sample validation using the V1 portfolio metrics. Additional tests such as walk-forward validation are proposed for future research.

Experimental results indicated the model's superiority through comparative baselines and ablation studies. With the help of the GA module and the dual MCDM layer, portfolio weight was optimized within the feasible space of GP to score and filter assets from various perspectives. As a result, the model generated better Sharpe ratios, reduced budget variances, and provided more transparent insights than conventional or single-model approaches. Finally, this integrated framework brings a modular, scalable and transparent method of portfolio construction, effectively bridging investor driven decision models and robust computational optimization.

It offered the TOPSIS-ARAS-GP-GA framework in the shape of multi-criteria/multi-objective portfolio optimization where asset scoring is qualitative whereas risk-return optimization is quantitative. The staged design had a good computational efficiency as evidenced by pre-optimization filtering and early convergence of the GA which made them scalable to large portfolios. The robustness was established through a-sensitivity examination (Kendall > 0.9) tests and resistance to moderate data quality concerns. Using benchmarking on Markowitz, NSGA-III, MOPSO, it was found that it was better or equalled in all four measures of Sharpe ratio, ROI, diversification, and budget adherence.

Module-based architecture and the automated processing in the back end made the approach practically applicable and available even to non-technical portfolio managers. The system can accomplish fast recalibration, as asset scores and investor profiles are refreshed with market data to enable periodic or issue-driven rebalancing. All of these features combined can provide stability, flexibility and usability within dynamic asset management settings, though future efforts are concentrated on dynamic rebalancing and hybrid metaheuristics to adapt more quickly.

Data availability
The datasets analysed during the current study are available in the following repository: [https://doi.org/10.5525/gla.researchdata.1658] .

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Author contributions

P. P and K.K. focused on the literature review, data processing, and model development. J. K supervised the study, while RRJ, MK, GB, and B. P. J. contributed to experimentation, result analysis, writing, and final proofreading.

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Declarations

Competing interests
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